© 2026 the authors. This work has been accepted to IFAC for publication under a Creative Commons Licence CC-BY-NC-ND.

Safety-Assured Arrival Scheduling in Sequential UAM Corridor Sections under Speed and Separation Constraints ⋆

Sasinee Pruekprasert * **Shinji Nakadai** ** **Katsuhiro Nishinari** *

* *The University of Tokyo, Tokyo, Japan. (e-mail: spruekprasert@g.ecc.u-tokyo.ac.jp, tknishi@mail.ecc.u-tokyo.ac.jp)*
** *Intent Exchange, Inc., Tokyo, Japan. (e-mail: nakadai@intent-exchange.com)*

Abstract: This paper presents a safety-assured arrival-scheduling framework for Urban Air Mobility (UAM) corridor operations. We propose an analytical method to compute a minimum safe Estimated Time of Arrival (ETA) gap at Constrained Waypoints (CWPs) that guarantees collision-free self-separation along sequential corridor sections with heterogeneous speed limits. The resulting ETA-gap condition depends on section-specific speed bounds and required separation distance, providing a closed-form rule suitable for integration into future digital ETA-scheduling and air traffic-management systems. We show that this sufficient ETA gap ensures safe separation across all corridor sections. Numerical simulations for decreasing-speed corridors confirm that vehicles coordinated with the proposed mechanism adjust their speeds to maintain the required spacing, avoid potential collisions, and support improved traffic flow compared to unscheduled operations.

Keywords: Urban Air Mobility, UAM corridors, Arrival scheduling, Safety assurance, Air traffic management

1. INTRODUCTION

Urban Air Mobility (UAM) introduces new air traffic management concepts for on-demand and scheduled aerial transportation in metropolitan areas, attracting growing attention from industry, academia, and government (Thipphavong et al., 2018). As demand increases, existing Air Traffic Control (ATC) systems may face capacity challenges under high traffic density (Vascik and Hansman, 2018). Moreover, while structured UAM corridors enhance safety, they have also been linked to increased travel delays compared with free-flight operations (Bauranov and Rakas, 2021). It is therefore essential to develop coordination mechanisms that maintain both safety and operational efficiency under these emerging constraints.

To address these challenges, the Federal Aviation Administration (FAA) incorporated UAM corridors into its UAM ConOps (Bradford, 2020; Fontaine, 2023). UAM corridors are three-dimensional reserved airspace volumes managed by designated authorities, such as Providers of Services for UAM (PSUs), who regulate access and performance. These corridors support applications such as air taxis and air ambulances (Muna et al., 2021), helping to reduce congestion and infrastructure constraints (Wang et al., 2021), and are envisioned primarily for electric Vertical Takeoff and Landing (eVTOL) and other powered-lift aircraft within the FAA's Advanced Air Mobility (AAM) framework.

Several prior studies have examined the design and management of UAM corridors. Some studies model corridors as structured graphs to analyze congestion and system-level performance (Wang et al., 2021; Jiang et al., 2022). Other studies examine geometric or layered architectures—such as cubes, skylanes, or vertically staggered tracks—and evaluate their safety characteristics (Muna et al., 2021). Broader taxonomies further highlight operational considerations and design trade-offs in structured airspace (Lee et al., 2023). Asslouj et al. (2024) extends structured-airspace concepts to fixed-corridor systems for Unmanned Aircraft Systems (UAS), offering scalable layouts and search-based path-assignment methods, but these approaches focus on corridor geometry and routing rather than deriving the temporal-separation conditions needed for safe inter-vehicle spacing, as envisioned under the Digital Flight operating mode (Wing et al., 2022).

Related research in conventional airspace management has explored coordinated scheduling and separation assurance at the system level. Integrated Demand Management in (Smith et al., 2016) combines strategic trajectory options with tactical time-based flow-management metering. Hildum and Smith (2012) formulates conflict-free airspace movement as a spatiotemporal scheduling problem. Human-in-the-loop studies such as (Wing et al., 2010) compare ground-based and airborne separation responsibility by evaluating pilot and controller performance in maintaining conflict-free trajectories. Pre-flight scheduling methods have also been proposed for advanced air mobility operations, such as cell-based demand–capacity balancing

⋆ This study is based on results obtained from the "Realization of Advanced Air Mobility (ReAMo) Project" by the New Energy and Industrial Technology Development Organization (NEDO).

and vertiport-resource allocation (Yokoyama et al., 2025), but these approaches focus on system-level planning rather than providing the temporal-separation constraints needed for safe spacing in corridor operations. Although these works highlight the value of scheduled-arrival management, they do not address the fine-grained, waypoint-level safety requirements for high-density UAM corridors.

Previous work (Pruekprasert and Nakadai, 2025) introduced an Estimated Time of Arrival (ETA)–based coordination scheme in which vehicles share their ETAs at constrained waypoints (CWPs) to ensure self-separation. By analytically deriving the minimum ETA gap required to satisfy Near Mid-Air Collision (NMAC) constraints (Johnson et al., 2017), the study provided formal safety guarantees for a single corridor section. These ETAs play a role similar to Required Times of Arrival (RTAs) used in trajectory-based self-separation and integrated flow-scheduling concepts (Wing et al., 2010; Smith et al., 2016), and align with emerging digital coordination modes such as Digital Flight (Wing et al., 2022), but tailored to UAM corridor operations. However, the single-section formulation does not extend to realistic UAM routes with multiple sections and differing speed limits, where safety must also hold when vehicles occupy different corridor sections.

Building on this foundation, this work extends the ETA-based safety framework to sequences of corridor sections, each with distinct speed bounds that may increase or decrease along the route. Such multi-section configurations naturally arise in realistic UAM operations due to changes in altitude layers, airspace policies, and local traffic densities. These heterogeneous speed limits introduce new challenges: a safe ETA gap computed for one section may no longer preserve separation after vehicles transition into an adjacent section with different constraints, and maintaining safety must account for the possibility that vehicles occupy different sections at the same time. To address these issues, we develop a scalable method for computing a *sufficient ETA gap* that guarantees safety across the entire sequence of corridor sections. Our method provides closed-form temporal-separation conditions that reduce the need for the complex optimization required in (Pruekprasert and Nakadai, 2025), while remaining scalable for long multi-section routes and suitable for integration into ETA-management systems for UAM corridors. We demonstrate how the proposed ETA-gap rule preserves safety along the corridor network and evaluate its efficiency through numerical simulations. We envision that the proposed framework can be incorporated into future digital ETA-management and ATC systems for UAM corridors.

The remainder of this paper is organized as follows. Section 2 introduces the UAM corridor setting and formulates the safe ETA-gap problem. Section 3 derives conservative spatiotemporal bounds under corridor speed limits, and Section 4 uses these bounds to construct a computationally efficient guaranteed-safe ETA gap. Section 5 presents numerical simulations. Section 6 presents the conclusion.

2. UAM CORRIDOR WITH ETAS AT CWPS

We consider Air Traffic Control (ATC) for inter-vehicle self-separation in Urban Air Mobility (UAM) corridors—structured airspace volumes envisioned for emerging AAM operations involving eVTOL and UAS platforms—using shared Estimated Times of Arrival (ETAs) at Constrained Waypoints (CWPs). As illustrated in Fig. 1, CWPs partition the airspace into multiple corridor sections. Each section may reserve specific airspace for different flight phases, and CWPs connect these sections to form an integrated UAM corridor network.

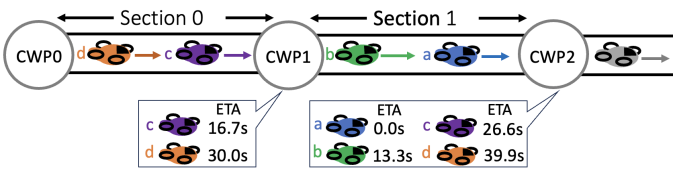

Fig. 1. A UAM corridor network with ETAs at CWPs.

2.1 Vehicle Kinematics and Corridor Constraints

We consider n aerial vehicles, indexed by $i \in \{0, \ldots, n-1\}$, entering the corridor network in order. Like (Pruekprasert and Nakadai, 2025), vehicle motion within each corridor section is modeled as one-dimensional and governed by

$$\dot{x}_i(t) = v_i(t), \qquad \dot{v}_i(t) = a_i(t), \tag{1}$$

where $x_i(t)$ and $v_i(t)$ denote the position and speed of vehicle i, and $a_i(t)$ is its acceleration input at time t.

For each corridor section j, we specify speed limits $v_{min,j}$ and $v_{max,j}$ representing the operational bounds imposed by the corridor design or airspace regulations. Accordingly, while vehicle i is within section j, its speed must satisfy

$$0 < v_{min,j} \leqslant v_i(t) \leqslant v_{max,j}. \tag{2}$$

To simplify ETA scheduling, we assign all vehicles a common travel time in each corridor section j to be $\tau_j = \frac{l_j}{v_{avg,j}}$, where $v_{avg,j} = (v_{min,j} + v_{max,j})/2$ is a representative average speed and l_j denotes the corridor section length. The proposed framework, however, can be extended to heterogeneous travel times.

2.2 Problem Formulation

According to Near Mid-Air Collision (NMAC) avoidance rules, a minimum separation distance is required to ensure safety between aerial vehicles, typically set to a fixed length (Johnson et al., 2017; Muna et al., 2021). We adopt this separation distance as the safety constraint for ETA-based scheduling at CWPs, requiring that the longitudinal spacing between vehicles never falls below *safeD*.

We consider m sequential corridor sections indexed by $j \in 0, \ldots, m-1$, where each section j connects CWP_j and CWP_{j+1}. Then, we formulate the problem of determining a safe ETA gap $\Delta t_{i,i+1}$ as follows.

Problem 1. For two consecutive vehicles i and $i+1$ entering the corridor network in order at CWP_0, find

$$\Delta t_{i,i+1} \;:=\; t_{i+1,\mathrm{enter}} - t_{i,\mathrm{enter}} \;> 0$$

$$\text{such that } x_i(t) - x_{i+1}(t) \;\geqslant\; safeD,$$

$$\text{for all } t \in [t_{i+1,\mathrm{enter}}, t_{i,\mathrm{enter}} + \tau_0 + \tau_1 + \cdots + \tau_{m-1}],$$

where $t_{i,\mathrm{enter}}$ and $t_{i+1,\mathrm{enter}}$ are the entry times at CWP_0 if vehicle i and $i+1$, respectively, and *safeD* is the NMAC minimum separation distance.

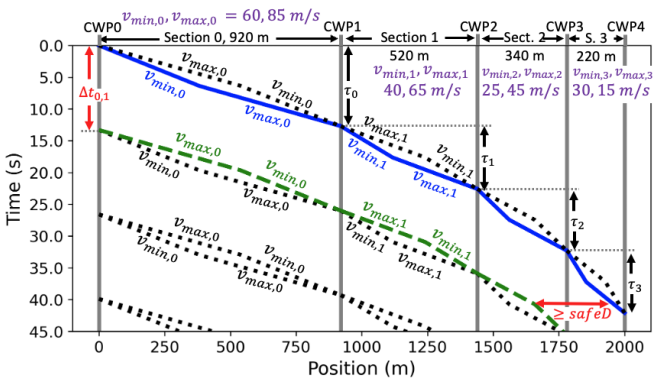

Fig. 2. Conservative spatiotemporal bounds of vehicles trajectories in UAM corridors with $\Delta t_{i,i+1} = 13.3$ s for *safeD* = 300 m. Length of corridors 0-3: 920, 520, 340, and 220 m; Speed limits: $v_{\min,0} = 60$, $v_{\max,0} = 85$; $v_{\min,1} = 40$, $v_{\max,1} = 65$; $v_{\min,2} = 25$, $v_{\max,2} = 45$; $v_{\min,3} = 15$, $v_{\max,3} = 30$ m/s. Travel times: $\tau_0 = 12.7$, $\tau_1 = 9.9$, $\tau_2 = 9.7$, $\tau_3 = 9.8$ s.

Note that because the travel times $\tau_0, \tau_1, \ldots, \tau_{m-1}$ are fixed, an ETA gap $\Delta t_{i,i+1}$ at CWP_0 also induces ETA gaps $\Delta t_{i,i+1} + \tau_0 + \tau_1 + \cdots + \tau_{j-1}$ at each CWP_j.

A trivial solution to Problem 1 is to choose $\Delta t_{i,i+1} > \tau_0 + \tau_1 + \cdots + \tau_{m-1}$, which ensures that vehicle $i+1$ enters the corridor at CWP_0 only after vehicle i has exited at CWP_m. For example, for the corridor sections in Fig. 2, this trivial solution requires $\Delta t_{i,i+1} > \tau_0 + \tau_1 + \tau_2 + \tau_3 = 42.1$ s. Although always safe, this choice is excessively conservative. We therefore aim for the smallest $\Delta t_{i,i+1}$ that still guarantees $x_i(t) - x_{i+1}(t) \geqslant safeD$ for all admissible trajectories.

In the next section, we propose a computationally efficient method to compute a sufficient ETA gap. Although this gap is not necessarily the globally minimal ETA gap, it is guaranteed to satisfy the safety condition in Problem 1 and is typically much smaller than the trivial bound above.

3. SPATIOTEMPORAL TRAJECTORY BOUNDS

To compute a sufficient ETA gap for Problem 1, we first construct conservative spatiotemporal bounds on vehicle trajectories, based on the corridor speed limits. These bounds enable a computationally efficient method for deriving a guaranteed-safe ETA gap in the next section.

Given that all vehicles operate within the speed range $[v_{min,j}, v_{max,j}]$ and must spend τ_j time in each corridor section j, their admissible spatiotemporal trajectories are bounded by the slopes shown in Fig. 2. These bounds arise from two extreme speed profiles: one that switches from $v_{min,j}$ to $v_{max,j}$ (blue solid line for vehicle 0 in Fig. 2) and one that switches from $v_{max,j}$ to $v_{min,j}$ (green dashed line for vehicle 1), assuming instantaneous speed changes.

Formally, let l_j denote the length of corridor section $j \in \{0, 1, \ldots, m-1\}$, and let CWP_0 be located at position 0. We define the cumulative time and position offsets as

$$L_j := \sum_{\ell=0}^{j-1} l_\ell \quad \text{and} \quad T_{i,j} := t_{i,\text{enter}} + \sum_{\ell=0}^{j-1} \tau_\ell, \tag{3}$$

for $j \in \{1, 2, \ldots, m\}$, with $L_0 = 0$ and $T_{i,0} = t_{i,\text{enter}}$. Hence, L_j is the position of CWP_j and $T_{i,j}$ is the scheduled time at which vehicle i arrives at CWP_j.

For each corridor section j, define $\check{t}_j$ as the unique nonnegative solution of

$$v_{\min,j}\, \check{t}_j + v_{\max,j}\big(\tau_j - \check{t}_j\big) = l_j, \tag{4}$$

representing the switching time of a trajectory that first flies at $v_{\min,j}$ and then at $v_{\max,j}$. Note that such a solution exists if $v_{\min,j}\tau_j \leqslant l_j \leqslant v_{\max,j}\tau_j$, which is assumed throughout. For $t \in [T_{i,j}, T_{i,j+1}]$, the trajectory $\check{x}_i(t)$ is

$$\check{x}_i(t) = \begin{cases} L_j + v_{\min,j}(t - T_{i,j}), \\ \qquad\qquad \text{if } T_{i,j} \leqslant t \leqslant T_{i,j} + \check{t}_j, \\ L_j + v_{\min,j}\check{t}_j + v_{\max,j}(t - T_{i,j} - \check{t}_j), \\ \qquad\qquad \text{if } T_{i,j} + \check{t}_j \leqslant t \leqslant T_{i,j+1}. \end{cases} \tag{5}$$

Similarly, define $\hat{t}_j$ as the unique nonnegative solution of

$$v_{\max,j}\, \hat{t}_j + v_{\min,j}\big(\tau_j - \hat{t}_j\big) = l_j, \tag{6}$$

corresponding to a trajectory that first flies at $v_{\max,j}$ and then $v_{\min,j}$. The trajectory $\hat{x}_i(t)$ is defined analogously by swapping $v_{\min,j}$ and $v_{\max,j}$, and replacing $\check{t}_j$ with $\hat{t}_j$ in (5).

We now show that these two trajectories bound all admissible motions of a vehicle that satisfy the speed limits (2) and the prescribed section travel times τ_j.

***Theorem** 1.* Consider any vehicle i that satisfies the speed limits condition (2), and spends τ_j time units in each corridor section $j \in \{0, \ldots, m-1\}$. Then, for all $t \in [T_{i,0}, T_{i,m}]$, the vehicle position $x_i(t)$ satisfies

$$\check{x}_i(t) \leqslant x_i(t) \leqslant \hat{x}_i(t),$$

i.e., every admissible trajectory lies between the two extreme spatiotemporal bounds.

Proof. Fix a corridor section j and consider any admissible trajectory $x_i(t)$ that satisfies the speed bounds (2) and the fixed travel time τ_j. Let t^* be any time such that vehicle i is flying in section j, and define $x^* := x_i(t^*)$. Then, we have $t^* \in [T_{i,j}, T_{i,j+1}]$ and $x^* \in [L_j, L_{j+1}]$. To prove the theorem, it suffices to show that x^* lies between the lower and upper bounds $\check{x}_i(t^*)$ and $\hat{x}_i(t^*)$.

Assume, for the sake of contradiction, that either $x^* < \check{x}_i(t^*)$ or $\hat{x}_i(t^*) < x^*$. We treat only the former case, as the latter follows symmetrically. Suppose $x^* < \check{x}_i(t^*)$, and let $\check{t}_j$ be the speed-switching time in (4).

If $t^* \leqslant T_{i,j} + \check{t}_j$, then from (5), we have $\check{x}_i(t^*) = L_j + v_{\min,j}(t^* - T_{i,j})$. Since $x^* < \check{x}_i(t^*)$, we obtain

$$\frac{x^* - L_j}{t^* - T_{i,j}} < \frac{\check{x}_i(t^*) - L_j}{t^* - T_{i,j}} = v_{\min,j},$$

implying an average speed strictly below $v_{\min,j}$ from $(L_j, T_{i,j})$ to (x^*, t^*), contradicting (2).

Otherwise, $t^* > T_{i,j} + \check{t}_j$. From the second case of (5),

$$\check{x}_i(t^*) = L_j + v_{\min,j}\check{t}_j + v_{\max,j}(t^* - T_{i,j} - \check{t}_j).$$

Using $L_{j+1} = L_j + v_{\min,j}\check{t}_j + v_{\max,j}(\tau_j - \check{t}_j)$ and $T_{i,j+1} = T_{i,j} + \tau_j$, this expression can be rewritten as

$$\check{x}_i(t^*) = L_{j+1} - v_{\max,j}(T_{i,j+1} - t^*).$$

Since $x^* < \check{x}_i(t^*)$, we obtain

$$L_{j+1} - x^* > L_{j+1} - \check{x}_i(t^*) = v_{\max,j}(T_{i,j+1} - t^*),$$

which requires an average speed strictly above $v_{\max,j}$ from (x^*, t^*) to $(L_{j+1},\, T_{i,j+1})$, contradicting (2).

As neither case is possible, $\check{x}_i(t^*) \leqslant x^* \leqslant \hat{x}_i(t^*)$ for all t^* in section j. Since T_j and L_j accumulate across sections, the bound holds over the entire corridor network. □

4. COMPUTING A SUFFICIENT ETA GAP

Using the conservative spatiotemporal bounds, we present a computationally efficient method for deriving a sufficient ETA gap. Our objective is to determine the smallest ETA gap that guarantees safe spacing for all admissible trajectories. Thereby, we consider the following problem.

***Problem** 2.* For two consecutive vehicles i and $i{+}1$ entering the corridor network in order, compute the minimum ETA gap $\Delta t_{i,i+1} := T_{i+1,0} - T_{i,0} \geqslant 0$ that satisfies

$$\check{x}_i(t) - \hat{x}_{i+1}(t) \;\geqslant\; safeD,$$

for all $t \in [T_{i+1,0}, T_{i,m}]$. Here, $T_{i,j}$ is the time that vehicle i arrives at CWP_j defined in (3), *safeD* is the NMAC minimum separation distance, and $\check{x}_i(t)$ and $\hat{x}_{i+1}(t)$ are the trajectory bounds defined in (5).

Following Theorem 1, we have the following Corollary.

***Corollary** 1.* Any $\Delta t_{i,i+1}$ that is a solution of Problem 2 is also a solution of Problem 1.

Next, we propose an algorithm to compute the minimum ETA gap for Problem 2 for each pair of vehicles i and $i{+}1$. To do so, we extract the *critical points* of the trajectory bounds $\check{x}_i(t)$ and $\hat{x}_{i+1}(t)$, which correspond to the times at which the slope changes. Let $\check{C}_i$ and $\hat{C}_{i+1}$ denote these sets of critical points, defined as follows:

$$\begin{aligned}\check{C}_i \;:=\; & \bigcup_{j=0}^{m-1} \Big\{ (L_j,\, T_{i,j}),\, (L_j + v_{\min,j}\,\check{t}_j,\, T_{i,j} + \check{t}_j), \\ & (L_{j+1},\, T_{i,j+1}) \Big\},\end{aligned}$$

$$\begin{aligned}\hat{C}_{i+1} \;:=\; & \bigcup_{j=0}^{m-1} \Big\{ (L_j,\, T_{i+1,j}),\, (L_j + v_{\max,j}\,\hat{t}_j, \\ & T_{i+1,j} + \hat{t}_j),\, (L_{j+1},\, T_{i+1,j+1}) \Big\},\end{aligned}$$

where $\check{t}_j$ and $\hat{t}_j$ are the unique nonnegative solutions of (4) and (6), respectively

For a given pair of vehicles i and $i{+}1$, we collect all time instants at which either trajectory bound changes slope. Define the set of relevant time instants as

$$\widetilde{T}_i := \{\, t : (x,t) \in \check{C}_i \text{ or } (x,t) \in \hat{C}_{i+1} \,\} \;\cap\; [T_{i+1,0},\, T_{i,m}].$$

The following theorem states that it is sufficient to consider distances between the trajectories bounds at the time instants in the set $\widetilde{T}_i$.

***Theorem** 2.* Given a pair of vehicles i and $i{+}1$ and a safety distance $safeD \geqslant 0$.

If $\check{x}_i(t) - \hat{x}_{i+1}(t) \;\geqslant\; safeD$ for all $t \in \widetilde{T}_i$,

Then $\check{x}_i(t) - \hat{x}_{i+1}(t) \;\geqslant\; safeD$ for all $t \in [T_{i+1,0},\, T_{i,m}]$.

Proof. Let $d(t) := \check{x}_i(t) - \hat{x}_{i+1}(t)$. From (5), both bounds $\check{x}_i(t)$ and $\hat{x}_{i+1}(t)$ are continuous piecewise-affine functions whose slope changes only at the critical times in $\widetilde{T}_i$. Hence, $d(t)$ is also continuous and affine on each interval between consecutive elements of $\widetilde{T}_i$.

Order the elements of $\widetilde{T}_i$ as $T_{i+1,0} = t_0 < t_1 < \cdots < t_K = T_{i,m}$. On each interval $[t_k, t_{k+1}]$, the function $d(t)$ is affine and thus attains its minimum at one of the endpoints. By the assumption of the theorem, $d(t_k) \geqslant safeD$ for all k, which implies $d(t) \geqslant safeD$ for all $t \in [t_k, t_{k+1}]$. Taking the union over all intervals yields $d(t) \geqslant safeD$ for all $t \in [T_{i+1,0},\, T_{i,m}]$, which proves the claim. □

By Theorem 2, it suffices to check the separation constraint only at the critical times $\widetilde{T}_i$. This leads to the following optimization problem, which can be solved efficiently by evaluating the constraints at $\widetilde{T}_i$.

***Problem** 3.* (ETA-Gap Optimization).

$$\begin{aligned}\min_{\Delta t_{i,i+1} \in [0,\, \sum_{j=0}^{m-1} \tau_j]} \quad & \Delta t_{i,i+1} \\ \text{subject to} \quad & \check{x}_i(t) - \hat{x}_{i+1}(t) \;\geqslant\; safeD, \quad \forall\, t \in \widetilde{T}_i, \\ & T_{i+1,0} - T_{i,0} \;=\; \Delta t_{i,i+1},\end{aligned}$$

***Remark** 1.* Since all vehicles are assumed to share the same travel time τ_j in each corridor section j, the resulting minimum ETA gap $\Delta t_{i,i+1}$ is the same for all $i \in \{0, \ldots, n{-}1\}$. We nevertheless formulate Problems 1, 2 and 3 for a generic pair $(i, i{+}1)$ to highlight that the proposed method naturally extends to heterogeneous section travel times τ_j across vehicles.

Combining the above results, Theorems 1 and 2 together with Corollary 1 yield the following conclusion.

***Corollary** 2.* Let $\Delta t_{i,i+1}$ be the solution of the Problem 3. Then $\Delta t_{i,i+1}$ also satisfies Problems 1 and 2. In particular, it guarantees

$$x_i(t) - x_{i+1}(t) \;\geqslant\; safeD \quad \text{for all } t \in [T_{i+1,0},\, T_{i,m}].$$

Thus, using this minimal ETA gap ensures that all vehicles remain at least *safeD* apart for all admissible trajectories.

***Remark** 2.* Corridor lengths and speed limits strongly affect the minimum ETA gap $\Delta t_{i,i+1}$. For instance, setting all corridor section lengths in Fig. 2 to 500 m increases the gap to 16.5 s (from 13.3 s in the current setting). We leave the optimization of corridor configuration for future work.

5. EXPERIMENTAL RESULT

5.1 Simulation Setting

We validate the theoretical results through numerical simulation, comparing operations with and without the proposed ETA gaps. We focus on a decreasing-speed corridor, which is particularly prone to traffic compression and collision risk as vehicles slow down (e.g., when approaching a vertiport), making it a challenging test case. All simulations are implemented in Python 3.12 with a discrete-time step of $\delta_t = 0.1\,\mathrm{s}$. We use the corridor illustrated in Fig. 2, with *safeD* values ranging from 100 m to 1200 m to assess performance under varying spacing requirements.

Vehicle motion follows the discrete-time version of (1):

$$x_i(k{+}1) = x_i(k) + \delta_t\, v_i(k), \quad v_i(k{+}1) = v_i(k) + \delta_t\, a_i(k),$$

where the acceleration is constrained by $a_i(k) \in [-3,\, 2]\,\mathrm{m/s^2}$.

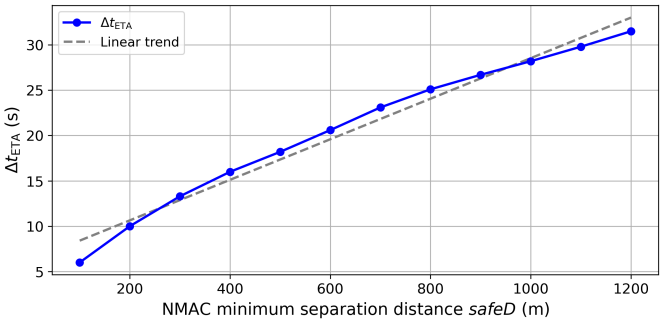

Fig. 3. Relationship between *safeD* and $t_{i,i+1}$.

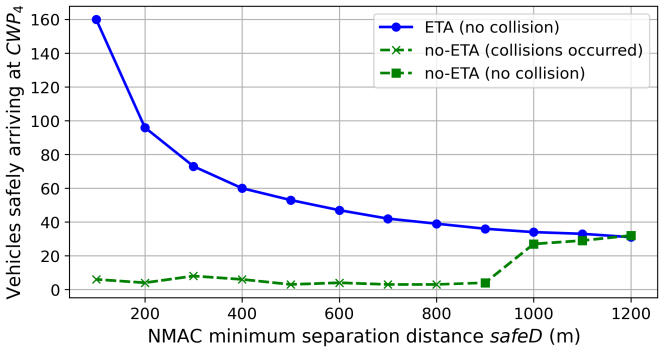

Fig. 4. Number of vehicles that safely arrived at CWP_4.

To model inter-vehicle interactions, we use the classical Helly car-following model (Helly, 1959; Ambrosio-Lázaro et al., 2018; Li et al., 2024). Accelerations depend on headway Δx, relative velocity Δv, and current speed v_k:

$$a_{\mathrm{Helly}} = \lambda_x(\Delta x - D(v_k)) + \lambda_v\,\Delta v, \tag{7}$$

where λ_x and λ_v are sensitivity coefficients, and

$$D(v_k) = D_{\min} + T_{\mathrm{des}} v_k$$

is the desired spacing, composed of a minimum distance $D_{\min}$ and an additional dynamic spacing $T_{\mathrm{des}} v_k$. We use the parameters $\lambda_x = 0.7$, $\lambda_v = 0.5$, $T_{\mathrm{des}} = 1.2$, and consider $D_{\min} = \mathit{safeD} \in \{100, 200, \ldots, 1200\}$ m. These parameter values except $D_{\min}$ follow (Li et al., 2024).

We consider continuous vehicle entry over a 100 s interval. All vehicles enter the network at CWP_0 with an initial speed of 82.5 m/s. Two operation modes are evaluated:

1) Without ETA gaps (no-ETA): Vehicles follow Helly's model (7), with accelerations clipped to $[-3, 2]\,\mathrm{m/s^2}$ and further adjusted to satisfy the section speed limits $v_{\min,j}$ and $v_{\max,j}$ of each corridor section j. A vehicle enters the corridor network at CWP_0 as soon as its computed Helly acceleration a_{Helly} becomes positive.

2) With ETA gaps (ETA): Vehicles also follow Helly's model under the same acceleration and speed-limit constraints, but the accelerations are additionally adjusted when necessary to ensure that the travel times match $\tau_0, \tau_1, \ldots, \tau_{m-1}$. Each vehicle enters at CWP_0 according to the schedule determined by the computed ETA gaps.

5.2 Simulation Results and Analysis

We solve the optimization Problem 3 for various values of *safeD*. Across all tested values, the minimum

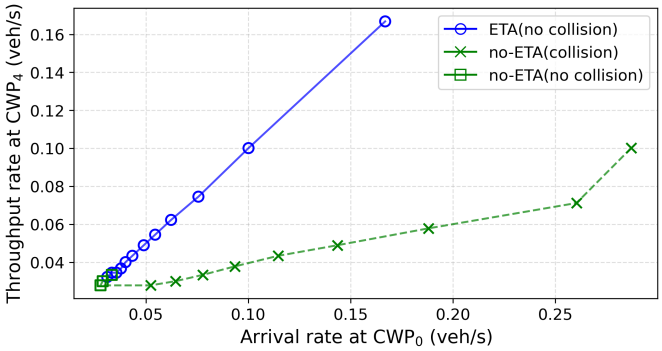

Fig. 5. Vehicle arrival rate at CWP_0 vesus the throughput rate at CWP_4.

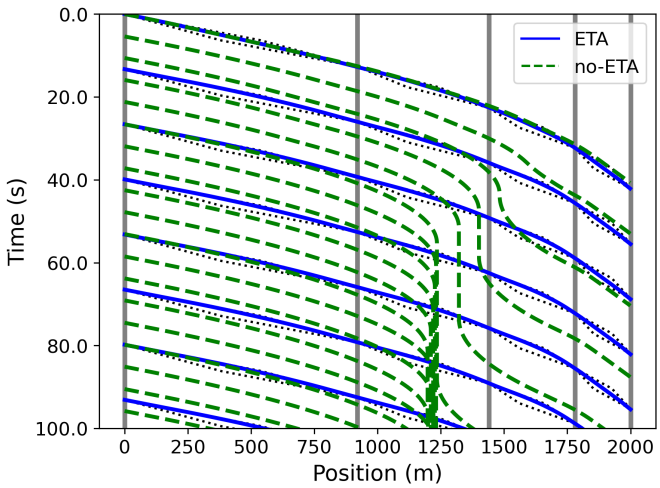

Fig. 6. Spatiotemporal trajectories of vehicles in UAM corridors for *safeD* = 300 m, under the two operation modes: with and without ETA gaps.

ETA gap $\Delta t_{i,i+1}$ is computed in under 0.003 s on a MacBook Pro (M4 Max chip) using the SciPy function `scipy.optimize.minimize_scalar`. The resulting ETA gaps for $\mathit{safeD} = 100, 200, \ldots, 1200$ m are 6.0, 10.0, 13.3, 16.0, 18.2, 20.6, 23.1, 25.1, 26.7, 28.2, 29.8, and 31.5 s, respectively. As shown in Fig. 3, the required ETA gap increases monotonically with *safeD*, with the rate of growth gradually diminishing for larger values.

Figure 4 shows the relation between *safeD* and the number of vehicles that successfully exit the corridor network at CWP_4 without collision. The ETA-gap policy prevents all collisions for the entire range of *safeD* considered, even for $\mathit{safeD} = 100$ m. In contrast, the no-ETA mode experiences collisions for all cases with $\mathit{safeD} \leqslant 800$ m. For the ETA-gap mode, the number of vehicles that complete the full corridor network within the simulation horizon (100 s) decreases as *safeD* increases, which is expected because larger separation distances require larger ETA gaps at entry. Nevertheless, for all $\mathit{safeD} \leqslant 1100$ m, the ETA-gap mode allows more vehicles to exit safely than the no-ETA mode. Figure 5 shows the arrival rate at CWP_0 versus the throughput rate at CWP_4. The ETA-gap mode achieves a nearly linear throughput increase, while the no-ETA mode saturates early and remains much lower. Thus, ETA-based coordination improves both safety and flow.

Moreover, across all simulations, every vehicle operating with ETA-gap remains at least *safeD* apart at all times.

Figure 6 shows the spatiotemporal trajectories for $safeD = 300\,\mathrm{m}$. Vehicles operating under the ETA-gap policy exhibit smooth, uniform flow: their trajectories follow the bounds $\check{x}_i(t)$ and $\hat{x}_i(t)$. In contrast, vehicles in the no-ETA mode experience reactive braking as they approach sections with reduced speed limits, resulting in congestion and, eventually, collisions.

These results show that the ETA-gap mechanism preserves safe inter-vehicle separation while substantially improving traffic flow compared to the unscheduled (no-ETA) mode.

6. CONCLUSION

We presented a computationally lightweight framework for safety-assured arrival scheduling in sequential UAM corridors. By coordinating ETAs at CWPs, we derived a minimal ETA-gap rule that guarantees inter-vehicle separation across corridor sections with heterogeneous speed limits using conservative spatiotemporal bounds. Numerical simulations confirmed that the ETA-gap policy maintained the required spacing in all scenarios, whereas unscheduled (no-ETA) operations experienced congestion and collisions. These findings demonstrate that ETA coordination can improve safety and flow stability in UAM corridors. For future work, we will consider additional separation criteria and corridor-design strategies.

REFERENCES

Ambrosio-Lázaro, R.C., Quezada-Téllez, L.A., and Rosas-Jaimes, O.A. (2018). Parameter identification on helly's car-following model. In *Proc. of 5th International Conference of Control, Dynamic Systems, and Robotics (CDSR'18)*.

Asslouj, A.E., Uppaluru, H., Ramezani, M., Atkins, E., and Rastgoftar, H. (2024). A fixed air corridor model for uas traffic management in urban areas. *IEEE Transactions on Aerospace and Electronic Systems*, 60(5), 5651–5662.

Bauranov, A. and Rakas, J. (2021). Designing airspace for urban air mobility: A review of concepts and approaches. *Progress in Aerospace Sciences*, 125, 100726.

Bradford, S. (2020). Urban air mobility concept of operations v1.0. Federal Aviation Administration, Office of NextGen. URL `https://nari.arc.nasa.gov/sites/default/files/attachments/UAM_ConOps_v1.0.pdf`. Accessed: 2025-11-20.

Fontaine, P. (2023). Urban air mobility concept of operations v2.0. Federal Aviation Administration, Office of NextGen. URL `https://www.faa.gov/sites/faa.gov/files/Urban%20Air%20Mobility%20%28UAM%29%20Concept%20of%20Operations%202.0_1.pdf`. Accessed: 2025-11-20.

Helly, W. (1959). Simulation of bottlenecks in single lane traffic flow. In *Proc. of Symposium on Theory of Traffic Flow*, 207–238. Elsevier, New York.

Hildum, D.W. and Smith, S.F. (2012). Scheduling safe movement of air traffic in crowded air spaces. *The Knowledge Engineering Review*, 27(3), 309–331.

Jiang, X., Peng, X., Bulusu, V., Poliziani, C., Chatterji, G., and Sengupta, R. (2022). A metrics-based method for evaluating corridors for urban air mobility operations. In *Proc. of 2022 IEEE International Smart Cities Conference (ISC2)*, 1–7. IEEE.

Johnson, S.C., Petzen, A., and Tokotch, D. (2017). Exploration of detect-and-avoid and well-clear requirements for small uas maneuvering in an urban environment. In *Proc. of 17th AIAA Aviation Technology, Integration, and Operations Conference*, 3074.

Lee, U.J., Ahn, S.J., Choi, D.Y., Chin, S.M., and Jang, D.S. (2023). Airspace designs and operations for UAS traffic management at low altitude. *Aerospace*, 10(9), 737.

Li, H., Roncoli, C., and Ju, Y. (2024). A helly model-based mpc control system for jam-absorption driving strategy against traffic waves in mixed traffic. *Applied Sciences*, 14(4), 1424.

Muna, S.I., Mukherjee, S., Namuduri, K., Compere, M., Akbas, M.I., Molnár, P., and Subramanian, R. (2021). Air corridors: Concept, design, simulation, and rules of engagement. *Sensors*, 21(22), 7536.

Pruekprasert, S. and Nakadai, S. (2025). Safe arrival scheduling at constraint waypoints in uam corridors. In *Proc. of AIAA SCITECH 2025 Forum*, 2232.

Smith, N.M., Brasil, C., Lee, P.U., Buckley, N., Gabriel, C., Mohlenbrink, C.P., Omar, F., Parke, B., Speridakos, C., and Yoo, H.S. (2016). Integrated demand management: Coordinating strategic and tactical flow scheduling operations. In *Proc. of 16th AIAA Aviation Technology, Integration, and Operations Conference*, 4221.

Thipphavong, D.P., Apaza, R., Barmore, B., Battiste, V., Burian, B., Dao, Q., Feary, M., Go, S., Goodrich, K.H., Homola, J., et al. (2018). Urban air mobility airspace integration concepts and considerations. In *Proc. of 2018 Aviation Technology, Integration, and Operations Conference*, 3676.

Vascik, P.D. and Hansman, R.J. (2018). Scaling constraints for urban air mobility operations: Air traffic control, ground infrastructure, and noise. In *Proc. of 2018 Aviation technology, Integration, and Operations conference*, 3849.

Wang, Z., Delahaye, D., Farges, J.L., and Alam, S. (2021). Air traffic assignment for intensive urban air mobility operations. *Journal of Aerospace Information Systems*, 18(11), 860–875.

Wing, D., Lacher, A., Ryan, W., Cotton, W., Stilwell, R., Maris, J., and Vajda, P. (2022). Digital flight: A new cooperative operating mode to complement vfr and ifr. URL `https://ntrs.nasa.gov/api/citations/20220013225/downloads/NASA-TM-20220013225.pdf`. Accessed: 2025-11-20.

Wing, D.J., Prevot, T., Murdoch, J.L., Cabrall, C.D., Homola, J.R., Martin, L.H., Mercer, J.S., Hoadley, S.T., Wilson, S.R., Hubbs, C.E., et al. (2010). Comparison of airborne and ground-based function allocation concepts for nextgen using human-in-the-loop simulations. In *Proc. of 10th AIAA Aviation Technology, Integration and Operations (ATIO) Conference*, ARC-E-DAA-TN1201.

Yokoyama, N., Shindo, M., Matayoshi, N., and Yoshida, H. (2025). Performance evaluation of uatm services accounting for airspace and vertiport capacities. In *Proc. of AIAA SCITECH 2025 Forum*, 2696.